

Musfirah Zainab

Backend Developer

Lahore, Pakistan

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About Me

A passionate Backend Developer with hands-on experience in building web and desktop applications using Python, FastAPI, C#, and JavaScript. Skilled in backend development, database integration, and application logic with a strong understanding of data structures and software architecture. Experienced in creating efficient and user-focused applications while continuously exploring modern backend technologies and development practices.

Technical Skills

- **Languages:** Python, C++, C#, JavaScript, HTML5, CSS3, MySQL
- **Frontend:** React.js, HTML5, CSS3, Vanilla JavaScript
- **Backend:** Python, Node.js, Express.js, ASP.NET, FastAPI
- **Desktop:** WinForms (C#, Visual Studio)
- **Databases:** SQL Server, PostgreSQL, MongoDB, Neo4j
- **Cloud:** Vercel
- **Concepts:** OOP, Data Structures & Algorithms, MVC Architecture
- **Tools:** Git, GitHub, Visual Studio, VS Code
- **MERN Stack:** MongoDB, Express.js, React.js, Node.js
- **Soft Skills:** Time Management, Teamwork, Communication, Problem Solving, Leadership

Projects

- | | | |
|--|---|--|
| Distributed Deadlock Detector | <i>Python, FastAPI, Distributed Systems</i> |  GitHub |
| <ul style="list-style-type: none">• Developed the backend of a distributed deadlock detection system using FastAPI.• Implemented APIs for process communication, resource allocation tracking, and deadlock detection handling.• Designed backend logic to simulate distributed system coordination and efficiently detect circular wait conditions. | | |
| Hardware Store Management System | <i>C#, .NET Framework, WinForms, SQL Server</i> |  GitHub |
| <ul style="list-style-type: none">• Built a C# WinForms application to manage inventory, sales, billing, and employee data for a hardware store.• Implemented role-based access control for Manager, Salesman, and Cashier roles.• Integrated database with full CRUD operations, discount logic, and reporting features. | | |
| Pharmacy Management System | <i>C#, .NET Framework, WinForms, SQL Server</i> |  GitHub |
| <ul style="list-style-type: none">• Developed a C# WinForms-based Pharmacy Management System to handle medicines, inventory, sales, billing, and customer records.• Implemented role-based access for Administrator, Pharmacist, and Cashier roles.• Integrated full database functionality including CRUD operations, expiry tracking, and automated billing. | | |
| Klondike Solitaire Card Game | <i>JavaScript, HTML5, CSS, Data Structures</i> |  GitHub |
| <ul style="list-style-type: none">• Developed a fully interactive browser-based Klondike Solitaire game using pure JavaScript with zero external libraries. | | |

- Implemented custom Stack, Queue, and LinkedList data structures for deck management, move tracking, and state handling.
- Built complete gameplay logic including drag-and-drop, draw-3 mechanics, Undo/Redo, scoring, timer, auto-move, and hints.

Question Paper Generator

HTML, CSS, JavaScript

Live Demo —  GitHub

- Developed a dynamic web application that automatically generates question papers from a question bank, ideal for educators and quiz creators.
- Implemented random generation logic to create unique question sets with every request.
- Built a clean, responsive user interface using vanilla JavaScript, HTML5, and CSS3 with no external frameworks.

Forks & Fiction – Restaurant Website

HTML, CSS, JavaScript

- Designed and developed a fully responsive restaurant website with real-time table booking functionality.
- Created an intuitive and visually appealing user interface with interactive features.
- Collaborated in a team environment using Git for version control and followed complete deployment workflows.

Education

University of Engineering and Technology (UET)

Bachelor of Science in Computer Science

CGPA: 3.7 / 4.0

Lahore, Pakistan

Expected 2028

Languages

- English: Professional Working Proficiency
- Urdu: Native
- Saraiki: Native